Power of a Multi-Crossover N-of-1 Design Versus a Parallel-Group Trial for a Treatment Main Effect: A Linear-DGP Pilot Simulation

pmsimstats team

October 28, 2026

Abstract

Background. Aggregated N-of-1 trials are proposed as an efficient alternative to parallel-group trials. Published simulations compare them at matched participant counts but not at matched observation counts. Carryover handling is rarely compared across rules that a trial can apply. **Methods.** We simulated a linear data-generating process for prazosin and PTSD nightmares. The N-of-1 arm was a randomized 4 + 4 multi-crossover of one-week periods in 35 participants. Five analysis rules handled carryover. Three parallel-group comparators used 35 participants. We derived power in closed form and checked it with 2,000 simulated replicates per cell. **Results.** The closed form gave power 0.84 for the schedule-exclusion rule and 0.24 for a one-visit trial at an effect of -1 nightmare per week. A trial with eight visits and a noisy baseline reached 0.63. The simulated powers were 0.83, 0.23, and 0.64, with Monte Carlo standard errors of at most 0.011. Null rejection rates ranged from 0.043 to 0.054 across arms. Ignoring carryover biased the estimate toward zero by 0.17 at an effect of -1. Under a true half-life of one week, schedule exclusion had simulated bias 0.19 and coverage 0.90. **Conclusions.** Most of the apparent N-of-1 advantage reflects the number of observations per participant. The remaining advantage is smaller and depends on how carryover is handled. This pilot does not simulate the hybrid design, treatment-effect heterogeneity, or a placebo response. **Keywords:** N-of-1 trials; randomized controlled trials; statistical power; carryover; crossover design; Monte Carlo simulation.

Contents

1 Introduction **2**

1.1 Scope of this paper . . . 3

1.2 Prior simulation comparisons . . . 3

1.3 Closed-form power . . . 3

1.4 Objectives . . . 4

2 Methods **4**

2.1 Protocol and deviations . . . 4

2.2 Aims . . . 4
2.3 Data-generating mechanisms . . . 5
2.4 Estimand . . . 6
2.5 Analysis methods . . . 6
2.6 Performance measures . . . 7
2.7 Closed-form performance . . . 7
2.8 Simulation size and random numbers . . . 7
2.9 Software . . . 8

3 Results **8**
3.1 Closed-form power . . . 8
3.2 Closed-form carryover results . . . 9
3.3 Agreement of the closed form with the stored 2026-05-19 run . . . 9
3.4 Simulation results . . . 10

4 Discussion **13**
4.1 Principal findings . . . 13
4.2 Carryover . . . 13
4.3 Comparison with prior studies . . . 14
4.4 Limitations . . . 14

5 Conclusions **14**

6 Glossary **15**

7 Abbreviations **15**

8 Declarations **16**

9 Reproducibility **16**

10 References **16**

1 Introduction

Aggregated N-of-1 trials combine several crossover series, one per participant, in a single analysis[1,2]. Each participant serves as the control. This removes between-participant variance

from the treatment contrast[3,4]. Simulation studies report that aggregated N-of-1 designs reach a given power with fewer participants than parallel-group trials[5,6].

This paper asks how much of that advantage is due to the design and how much to the number of observations per participant. It also asks how the choice of carryover rule changes the answer. The setting is prazosin for trauma-related nightmares in PTSD. The first randomized trials of prazosin were positive[7,8]. A later larger trial in veterans found no difference from placebo on its primary outcomes[9]. The prior evidence for the effect size is therefore mixed. The effect sizes in this paper are chosen for illustration.

1.1 Scope of this paper

This paper is a pilot. It uses a linear data-generating process (DGP) and a randomized multi-crossover design. It does not simulate the Hybrid design of Hendrickson et al.[10], treatment-effect heterogeneity, a placebo response, or a natural-history trend. The working protocol for a fuller study (`00-ademp-protocol.md`) describes those components. That study has not been run. The protocol that governs the present pilot is `00-ademp-protocol-v1.md`. The section on deviations lists every difference between that protocol and what was done.

The study is exploratory. No hypothesis in it was specified before the data were generated, and its findings need confirmation in an independent study.

1.2 Prior simulation comparisons

Table 1 summarizes the closest published simulations. Blackston et al. report 92% power for aggregated N-of-1 trials, 32% for a parallel-group trial, and 50% for a two-period crossover at $n = 30$ in their first variance scenario[5]. Their parallel-group trial also supplies one observation per participant. The observation-count difference is therefore present in their design as well. Percha et al. simulate N-of-1 series with carryover, wash-in, and autocorrelation, and give design recommendations on block count, order, period length, and sampling frequency[11]. Wang and Schork give analytical power under AR(1) errors, heteroscedasticity, washout, and carryover[12].

Table 1: Closest published simulation and analytical studies.

Study	Design	Focus
Blackston et al. 2019	Aggregated N-of-1, parallel, crossover	Power, Type I error, carryover
Chen and Chen 2014	Four analyses of aggregated N-of-1	Type I error and power by analysis, carryover regressor
Percha et al. 2019	Stochastic N-of-1 series simulator	Block count, order, period length, sampling
Wang and Schork 2019	Crossover N-of-1, analytical power	AR(1) errors, washout, carryover
Carrozzo et al. 2025	Parallel, crossover, N-of-1 meta-analysis	Heterogeneity, carryover, sequential aggregation
Hendrickson et al. 2020	Open label, BDC, crossover, Hybrid	Biomarker-treatment interaction

1.3 Closed-form power

Senn gives the variance of the mean treatment difference for a series of paired cycles as $(\tau^2 + 2\sigma^2/k)/n$ for n participants and k cycles[13]. Here τ^2 is the between-participant variance

of the drug effect and σ^2 is the within-period variance. Yang et al. extend closed-form planning for N-of-1 series[14]. The DGP in this paper has $\tau^2 = 0$. The closed form therefore applies directly, and the Methods derive it. The simulation confirms it and checks the finite-sample behavior of the inference.

1.4 Objectives

The objectives were fourfold.

1. To derive the power of each analysis arm in closed form.
2. To compare a multi-crossover N-of-1 design with parallel-group comparators of one visit and of eight visits, at 35 participants.
3. To compare five carryover rules, including one that uses the truth and is labeled as such.
4. To check the closed form and the Type I error by Monte Carlo simulation.

2 Methods

We follow the ADEMP structure of Morris et al.[15].

2.1 Protocol and deviations

The governing protocol is the version 1 working protocol, `00-ademp-protocol-v1.md`, in `analysis/scripts/treatment-main-effect/`. Table 2 records each deviation from it. The rebuild protocol of October 2026 is a separate study and was not run.

Table 2: Deviations from the version 1 protocol.

Item	Version 1 protocol	This pilot
N-of-1 design	Hybrid (open label, BDC, crossover)	Randomized 4 + 4 multi-crossover, one-week periods
Outcome model	Three-component Gompertz DGP	Linear DGP, no heterogeneity, trend, or placebo response
RCT design	16 weeks, visits at weeks 0, 4, 8, 12, 16	One visit, or eight weekly visits
Sweeps	Twelve sweeps (S1 to S12)	M1 effect sweep and S1 half-life sweep
Carryover scale	Days, half-life 3 days	Weeks, primary half-life 0.2 week
Carryover handling	Not specified	Five rules, one labeled oracle
Inference	Wald t-test	Containment df and CR2 with Satterthwaite df
Seeding	One seed, one stream	One stream per replicate, stored
Deviation log	Promised, not written	This table

2.2 Aims

The aim was to quantify power, bias, coverage, and Type I error for a treatment main effect in a multi-crossover N-of-1 design and in parallel-group comparators, and to show how the carryover rule changes them.

2.3 Data-generating mechanisms

Units. Time is in weeks. One period is one week. The carryover half-life $t_{1/2}$ is in weeks. One week is seven days.

Each replicate draws $N = 35$ participant intercepts $u_i \sim \mathcal{N}(0, \sigma_u^2)$ with $\sigma_u = 1.5$. The same participants enter every arm of that replicate. The baseline mean is $\mu_0 = 8$ nightmares per week.

In the N-of-1 arm, participant i completes $T = 8$ weekly periods, four on drug and four on placebo, in a random order. The outcome of period t is

$$Y_{it} = \mu_0 + u_i + \beta_D D_{it} + \beta_D \, c_{it} + \varepsilon_{it}.$$

Here D_{it} is the drug indicator and β_D is the on-drug effect. The residuals ε_{it} have standard deviation $\sigma_\varepsilon = 2.3$ and lag-one correlation $\rho = 0.3$ per week, so that $\mathrm{Cor}(\varepsilon_{it}, \varepsilon_{is}) = \rho^{|w_t - w_s|}$. The treatment-effect heterogeneity is $\tau^2 = 0$. The intraclass correlation of one measurement is $\sigma_u^2/(\sigma_u^2 + \sigma_\varepsilon^2) = 0.30$.

The carryover term c_{it} is zero on drug and before first exposure. In an off-drug period after exposure it is the mean of $2^{-s/t_{1/2}}$ over the period,

$$c_{it} = \frac{t_{1/2}}{\ln 2} \left(2^{-k_{it}/t_{1/2}} - 2^{-(k_{it}+1)/t_{1/2}} \right),$$

where k_{it} is the number of completed off-drug periods since the last drug period. The carryover decays from the last on-drug mean and continues across consecutive off-drug periods. A half-life of zero means no carryover.

The primary half-life is 0.2 week, inside the pharmacological range for prazosin. Its plasma half-life of 2 to 3 hours[16] is shorter, so 0.2 week is a conservative allowance for pharmacodynamic persistence. Longer half-lives of 0.5, 1, and 2 weeks are behavioral scenarios. They are not pharmacological estimates.

Table 3 lists the parameters. The values of μ_0, σ_u, σ_ε, ρ, and the effect grid are assumptions chosen to be of plausible size. They are not estimated from Raskind et al.[8] or Hendrickson et al.[10]. The first of these measured nightmares with the Clinician-Administered PTSD Scale dream item and not with weekly counts.

Parallel-group comparators. The same 35 participants are allocated 17 to 18 at random. Three comparators differ in visits and in the baseline covariate.

- RCT1_exact: one post-baseline visit, ANCOVA on the true intercept u_i. This is the comparator of the earlier version of this paper.
- RCT1_r08: one visit, ANCOVA on a baseline covariate with reliability $r_{BL} = 0.8$.
- RCT8_r08: eight weekly visits, ANCOVA on the participant mean of the eight visits, with the same covariate. For balanced complete data this equals the generalized least

Table 3: Simulation parameters and their basis.

Parameter	Symbol	Value	Basis
Participants N	N	35	Matches the version 1 protocol
Periods T	T	8	Assumed
Baseline mean	μ_0	8 nightmares/week	Assumed
Between-participant SD	σ_u	1.5	Assumed
Residual SD	σ_ε	2.3	Assumed
Lag-one correlation	ρ	0.3/week	Assumed
Effect grid	β_D	0, -0.5, -1, -2 /week	Assumed, illustrative
Primary half-life	$t_{1/2}$	0.2 week	Pharmacological range
Half-life sweep	$t_{1/2}$	0, 0.2, 0.5, 1, 2 weeks	Behavioral scenarios
Analysis half-life	$t^a_{1/2}$	0.2 week	Policy value (NOTATION.md)
Baseline reliability	r_{BL}	1, 0.8	1 is error-free, 0.8 is a stated scenario

squares estimator of the mixed model. It matches the N-of-1 arm in the number of observations per participant.

A covariate with reliability r_{BL} is u_i plus independent noise of variance $\sigma_u^2(1 - r_{BL})/r_{BL}$. A single baseline visit from this DGP would have reliability equal to the intraclass correlation above. The value 0.8 is therefore a favorable scenario for the parallel-group trial.

2.4 Estimand

The estimand is the on-drug effect β_D in nightmares per week. In the N-of-1 arm it is the direct effect of drug during an on-drug period. Power is the probability that the two-sided test of $\beta_D = 0$ rejects at the 0.05 level.

2.5 Analysis methods

Each N-of-1 rule fits

$$Y_{it} = \beta_0 + \beta_D\, x_{it} + \gamma\, L_{it} + \beta_T\, w_t + u_i + \varepsilon_{it}$$

with a random intercept and a continuous-time AR(1) residual correlation (`nlme::lme`, `corCAR1`, REML). The term γL_{it} enters only in the DBCL rule. Table 4 defines the five rules.

Table 4: N-of-1 analysis rules.

Rule	Regressor	Uses the truth
ALL	Binary D, all periods	No
XS	Binary D, first off-drug period after each drug period dropped	No
DBC	D_{bc} at $t^a_{1/2} = 0.2$ week, all periods	No (assumes 0.2)
DBCL	D_{bc} at $t^a_{1/2} = 0.2$ week plus lag indicator L, all periods	No (assumes 0.2)
XO	Binary D, off-drug periods with true carryover above 0.05 dropped	Yes (oracle)

The rule XS uses the schedule only and is applied identically in the null and alternative cells. The rule XO is an oracle. It uses the true half-life, and it is reported only to show what perfect knowledge would give. The rule D_{bc} is the drug indicator in on-drug periods and the carryover fraction at $t_{1/2}^{a}$ in off-drug periods after exposure. It is zero before first exposure.

Inference uses two methods for each rule. The first is the `nlme` Wald t-test with containment (between-within) degrees of freedom. The second is CR2 cluster-robust standard errors with Satterthwaite degrees of freedom[17]. The `nlme` package has no Satterthwaite option. The earlier version of this paper called its containment-df test Satterthwaite. That description was wrong.

2.6 Performance measures

For every arm and cell we report power, bias, empirical standard error, mean model standard error, and 95% interval coverage, each with its Monte Carlo standard error (MCSE). The MCSE of a proportion $\hat{p}$ is $\sqrt{\hat{p}(1-\hat{p})/n_{\text{sim}}}$. The paired difference in power between two arms uses the replicate-level rejection indicators. Its MCSE is the standard deviation of the indicator difference divided by $\sqrt{n_{\text{sim}}}$.

2.7 Closed-form performance

Under the DGP, the covariance of one participant's outcomes is $V = \sigma_u^2 \mathbf{1}\mathbf{1}^\top + \sigma_\varepsilon^2 R(\rho)$, with $R_{ts} = \rho^{|t-s|}$. For a rule with retained rows and regressor matrix X_i, the generalized least squares variance of $\hat{\beta}_D$ is the β_D element of $(\sum_i X_i^\top V_i^{-1} X_i)^{-1}$. We average the information over the 70 equally likely balanced sequences. The expected estimate is the same inverse times $\sum_i X_i^\top V_i^{-1} \mu_i$, where μ_i is the true mean including carryover. The difference from β_D is the bias. Power and coverage follow from the noncentral t distribution with the containment degrees of freedom and noncentrality $(\beta_D + \text{bias})/\text{SE}$.

For the parallel-group comparators, the residual variance after covariate adjustment is $\sigma_\varepsilon^2\, s + \sigma_u^2(1 - r_{BL})$, with $s = \mathbf{1}^\top R \mathbf{1}/m^2$ for m visits. The variance of the effect is that residual variance times $1/17 + 1/18$.

The closed form ignores the sampling error of the estimated variance components. The simulation measures that error.

2.8 Simulation size and random numbers

Each cell has $n_{\text{sim}} = 2{,}000$ replicates. At this size the MCSE of a power near 0.75 is 0.010, and of a rejection rate near 0.05 it is 0.005. There are two sweeps. The M1 effect sweep has $\beta_D \in \{0, -0.5, -1, -2\}$ at $t_{1/2} = 0.2$. The S1 half-life sweep has $t_{1/2} \in \{0, 0.5, 1, 2\}$ at $\beta_D = -1$. The carryover term is proportional to β_D, so a null cell does not depend on $t_{1/2}$.

One L'Ecuyer-CMRG chain, advanced with `parallel::nextRNGStream()`, gives every replicate in every cell its own stream. The streams are stored, and results do not depend on the

number of cores. This replaces the earlier driver, in which `mclapply` reused one set of streams in all nine cells.

2.9 Software

The analyses use R 4.4 or later with `nlme`, `clubSandwich`, and `parallel`. The code is in `analysis/scripts/treatment-main-effect/`. The file `pilot-core.R` holds the DGP, the analysis rules, and the closed form. The file `production-run.R` is the driver, and `check-pilot.R` runs the fast checks. Package versions are in `renv.lock`.

3 Results

3.1 Closed-form power

Table 5: Closed-form power at the primary half-life (0.2 week), 35 participants. XO is an oracle.

Arm	$\beta_D = -0.5$	$\beta_D = -1$	$\beta_D = -2$
XS	0.313	0.837	1.000
DBC	0.407	0.932	1.000
DBCL	0.324	0.852	1.000
XO	0.313	0.837	1.000
RCT1_exact	0.096	0.239	0.703
RCT1_r08	0.092	0.224	0.668
RCT8_r08	0.208	0.627	0.995

At $\beta_D = -1$ nightmare per week, the closed-form power of the schedule-exclusion rule XS is 0.84. The one-visit comparator with an error-free covariate has power 0.24. With a covariate of reliability 0.8 it has 0.22. The eight-visit comparator has 0.63. The rule DBC has 0.93. Figure 1 shows the power curves.

Table 6: Closed-form standard error of the effect estimate and mean analyzed observations per participant.

Arm	SE of the effect	Analyzed obs. per participant	SE relative to XS
XS	0.336	6	1.00
DBC	0.289	8	0.86
DBCL	0.332	8	0.99
XO	0.336	6	1.00
RCT1_exact	0.778	1	2.31
RCT1_r08	0.810	1	2.41
RCT8_r08	0.425	8	1.26

Table 6 separates the sources of the difference. The standard error of RCT1_exact is 2.31 times that of XS. The standard error of RCT8_r08 is 1.26 times that of XS. Giving the parallel-group trial eight visits removes most of the standard-error gap. The N-of-1 arm

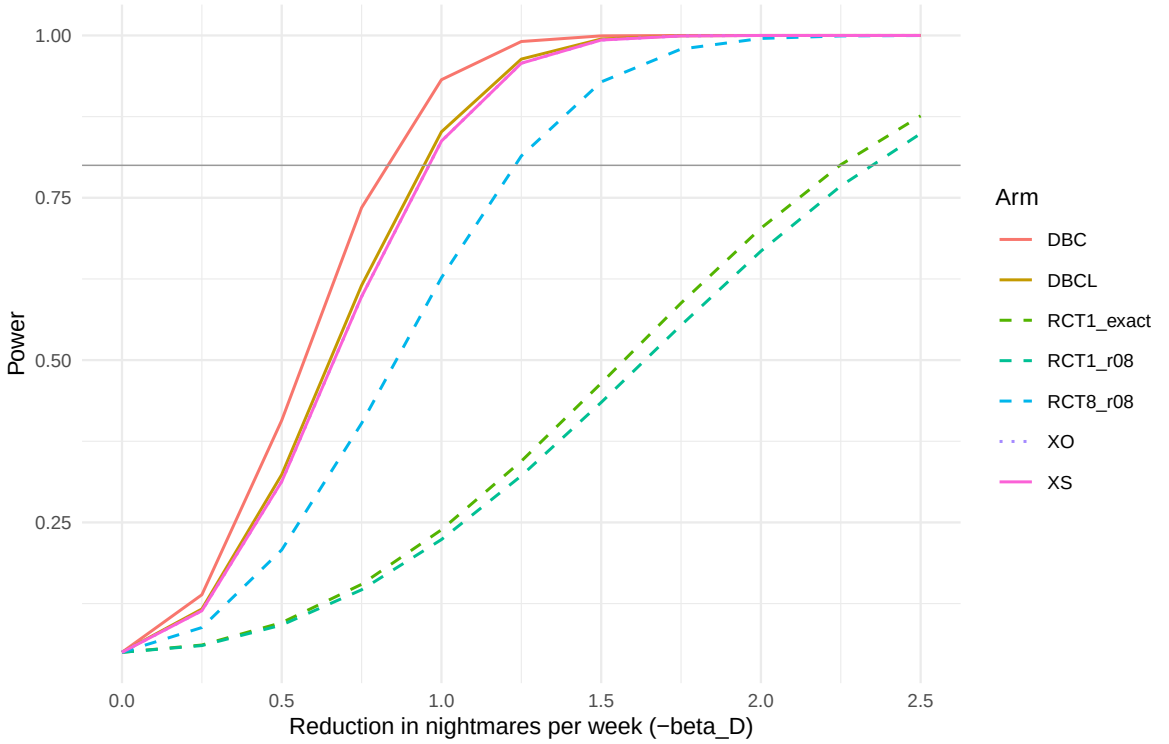


Figure 1: Closed-form power against the true effect at a half-life of 0.2 week. The dotted line is the oracle rule XO, which a trial cannot apply.

analyzes six periods per participant under XS and eight under DBC. The rule DBC has the smaller standard error of the two because it retains the periods that XS drops.

3.2 Closed-form carryover results

Table 7 and Figure 2 show how the rules behave as the true half-life grows. At 0.2 week, XS has bias 0.004 and DBC has none, because DBC is built at the true half-life. At 0.5 week, XS has bias 0.071 and DBC has 0.174. At one week, XS has bias 0.20 and coverage 0.91. The rules DBC and DBCL assume 0.2 week. When the true half-life is zero, DBC has bias -0.148, because it adjusts for carryover that does not exist.

The oracle XO is unbiased at every half-life. It pays for this with analyzed periods. At one week, XO analyzes 4.8 periods and has power 0.41, against 0.65 for XS. No rule that a trial can apply is both unbiased and well powered at a half-life of one week or longer without a longer washout.

3.3 Agreement of the closed form with the stored 2026-05-19 run

The earlier version of this paper stored a simulation of a related design. In that run the null cell analyzed all eight periods, and the alternative cells analyzed about six, because exclusion used a flag that was nonzero only when $\beta_D \neq 0$. The closed form reproduces its standard

Table 7: Closed-form bias, power, and 95% interval coverage by rule and true half-life at $\beta_D = -1$. Periods is the mean number analyzed per participant. XO is an oracle.

$t_{1/2}$	Rule	Bias	Power	Coverage	Periods
0.0	ALL	0.000	0.982	0.950	8.0
0.0	XS	0.000	0.840	0.950	6.0
0.0	DBC	-0.148	0.977	0.920	8.0
0.0	DBCL	-0.004	0.855	0.950	8.0
0.0	XO	0.000	0.982	0.950	8.0
0.2	ALL	0.170	0.920	0.894	8.0
0.2	XS	0.004	0.837	0.950	6.0
0.2	DBC	0.000	0.932	0.950	8.0
0.2	DBCL	0.000	0.852	0.950	8.0
0.2	XO	0.004	0.837	0.950	6.0
0.5	ALL	0.349	0.752	0.706	8.0
0.5	XS	0.071	0.784	0.945	6.0
0.5	DBC	0.174	0.813	0.908	8.0
0.5	DBCL	0.068	0.799	0.945	8.0
0.5	XO	0.012	0.598	0.950	5.1
1.0	ALL	0.503	0.523	0.469	8.0
1.0	XS	0.203	0.654	0.908	6.0
1.0	DBC	0.345	0.618	0.779	8.0
1.0	DBCL	0.203	0.668	0.907	8.0
1.0	XO	0.000	0.405	0.950	4.8
2.0	ALL	0.630	0.323	0.275	8.0
2.0	XS	0.356	0.477	0.816	6.0
2.0	DBC	0.499	0.409	0.594	8.0
2.0	DBCL	0.359	0.486	0.810	8.0
2.0	XO	0.000	0.405	0.950	4.8

errors and power (Table 8). The comparison is a check on the closed form. It does not validate the v1 results, which are superseded. The stored RCT null rejection rate of 0.062 is 2.1 MCSE above 0.05. The ANCOVA test is exact under Gaussian errors, so the excess is most likely chance in a run whose cells shared one set of random numbers.

Table 8: Closed form against the stored 2026-05-19 run (2,000 replicates per cell, one shared random-number stream).

β_D	N-of-1 SE, stored	N-of-1 SE, closed form	N-of-1 power, stored	N-of-1 power, closed form	RCT power, stored	RCT power, closed form
0.0	0.246	0.245	0.051	0.050	0.062	0.050
-0.5	0.342	0.336	0.312	0.313	0.104	0.096
-1.0	0.342	0.336	0.830	0.837	0.230	0.239
-2.0	0.342	0.336	1.000	1.000	0.675	0.703

3.4 Simulation results

We ran 2,000 replicates in each of the eight cells, seeded as described under Reproducibility. Of the 128,000 model fits, 7 failed. Every failure was in XS or XO, and each cell retained at least 1,997 replicates per arm. Table 9 gives the rejection rates of the M1 effect sweep.

The Type I error ranged from 0.043 to 0.054 across the eight arms with containment-df inference. The MCSE of each rate is about 0.005. With CR2 and Satterthwaite df, the N-of-1 rules ranged from 0.048 to 0.057. No rate differs from 0.05 by more than 1.5 MCSE.

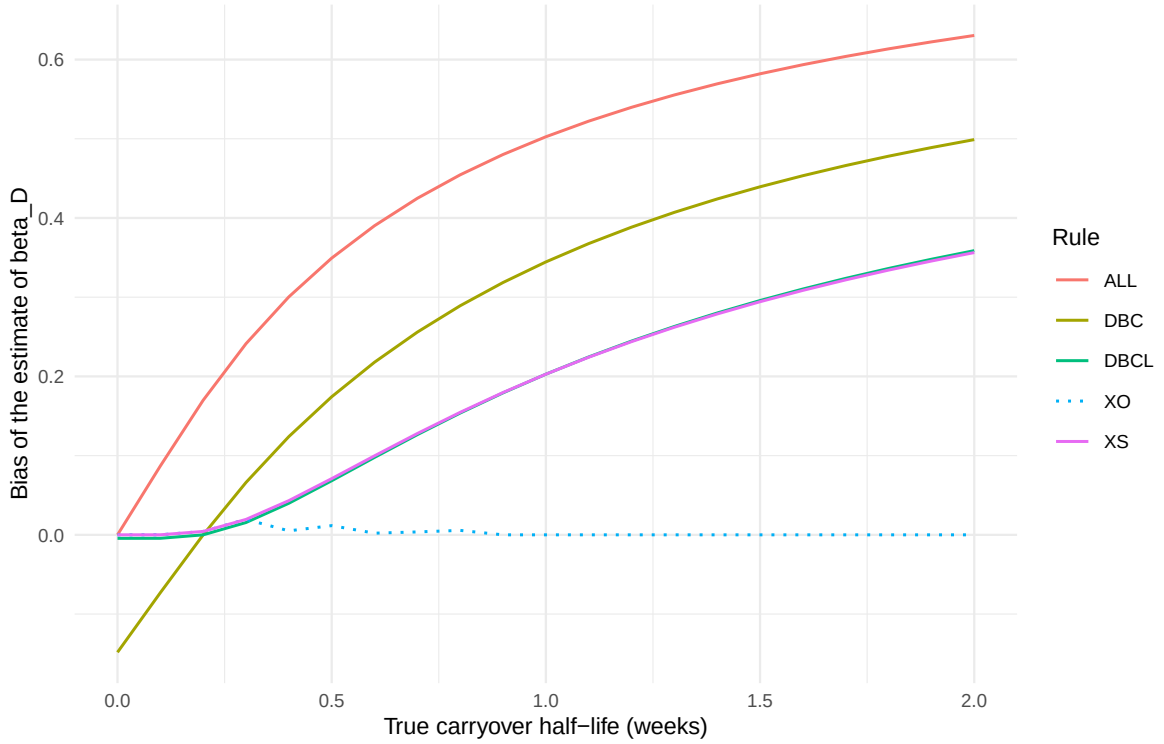


Figure 2: Closed-form bias of the effect estimate against the true half-life at a true effect of -1 nightmare per week. The dotted line is the oracle rule XO.

Table 9: Simulated rejection rate (MCSE) with containment-df inference at a half-life of 0.2 week. The column for $\beta_D = 0$ is the Type I error. XO is an oracle.

Arm	$\beta_D = 0$	$\beta_D = -0.5$	$\beta_D = -1$	$\beta_D = -2$
ALL	0.047 (0.005)	0.399 (0.011)	0.915 (0.006)	1.000 (0.000)
XS	0.053 (0.005)	0.319 (0.010)	0.826 (0.008)	1.000 (0.000)
DBC	0.051 (0.005)	0.412 (0.011)	0.929 (0.006)	1.000 (0.000)
DBCL	0.048 (0.005)	0.324 (0.010)	0.842 (0.008)	1.000 (0.000)
XO	0.053 (0.005)	0.319 (0.010)	0.826 (0.008)	1.000 (0.000)
RCT1_exact	0.054 (0.005)	0.093 (0.006)	0.232 (0.009)	0.679 (0.010)
RCT1_r08	0.043 (0.005)	0.088 (0.006)	0.218 (0.009)	0.664 (0.011)
RCT8_r08	0.052 (0.005)	0.203 (0.009)	0.637 (0.011)	0.995 (0.002)

XS and XO gave identical estimates in every M1 replicate. At a half-life of 0.2 week, the carryover fraction is 0.280 in the first off-drug period and 0.0087 in the second. The oracle cut is 0.05, so XO drops exactly the periods that XS drops by schedule.

Without carryover handling, ALL was biased toward zero. Its bias was 0.084 at $\beta_D = -0.5$, 0.172 at -1, and 0.346 at -2, with MCSE 0.006. Its coverage at the last two effects was 0.89 and 0.70. The bias of XS, DBC, and DBCL was at most 0.018 in absolute value across the M1 cells.

Table 10 compares the simulation with the closed form at $\beta_D = -1$. The simulated power was 0.826 for XS, 0.929 for DBC, 0.232 for RCT1_exact, and 0.637 for RCT8_r08. The closed-

Table 10: Simulation against closed form at $\beta_D = -1$ and half-life 0.2 week. The column z is the simulated minus the closed-form power, divided by the MCSE.

Arm	Power, sim.	MCSE	Mean model SE	Empirical SE	Power, closed form	SE, closed form	z
ALL	0.915	0.006	0.245	0.252	0.920	0.245	-0.788
XS	0.826	0.008	0.335	0.338	0.837	0.336	-1.334
DBC	0.929	0.006	0.288	0.295	0.932	0.289	-0.493
DBCL	0.842	0.008	0.331	0.332	0.852	0.332	-1.174
XO	0.826	0.008	0.335	0.338	0.837	0.336	-1.334
RCT1_exact	0.232	0.009	0.787	0.802	0.239	0.778	-0.759
RCT1_r08	0.218	0.009	0.820	0.816	0.224	0.810	-0.555
RCT8_r08	0.637	0.011	0.428	0.433	0.627	0.425	0.985

form values were 0.84, 0.93, 0.24, and 0.63. Every arm was within 1.3 MCSE of the closed form. The simulated power of each N-of-1 rule was below the closed form, by 0.003 to 0.011. The five rules are fitted to the same data, so these differences are not independent. Their direction is consistent with the sampling error of the variance components, which the closed form ignores. The mean model standard error was within 1.2% of the closed-form standard error for every arm.

Table 11 gives the S1 half-life sweep at $\beta_D = -1$, with the M1 cell at 0.2 week included.

Table 11: Simulated bias, power, and 95% interval coverage by rule and true half-life at $\beta_D = -1$, containment-df inference. The MCSE of power and coverage is at most 0.012. Periods is the mean number analyzed per participant. XO is an oracle.

$t_{1/2}$	Rule	Bias	MCSE of bias	Power	Coverage	Periods
0.0	ALL	-0.005	0.006	0.981	0.946	8.0
0.0	XS	-0.016	0.008	0.857	0.950	6.0
0.0	DBC	-0.158	0.007	0.975	0.918	8.0
0.0	DBCL	-0.022	0.007	0.871	0.956	8.0
0.0	XO	-0.005	0.006	0.981	0.946	8.0
0.2	ALL	0.172	0.006	0.915	0.892	8.0
0.2	XS	0.005	0.008	0.826	0.957	6.0
0.2	DBC	0.003	0.007	0.929	0.950	8.0
0.2	DBCL	0.001	0.007	0.842	0.956	8.0
0.2	XO	0.005	0.008	0.826	0.957	6.0
0.5	ALL	0.352	0.006	0.747	0.700	8.0
0.5	XS	0.072	0.008	0.770	0.949	6.0
0.5	DBC	0.177	0.007	0.810	0.901	8.0
0.5	DBCL	0.068	0.008	0.788	0.952	8.0
0.5	XO	0.006	0.010	0.603	0.951	5.1
1.0	ALL	0.493	0.006	0.531	0.488	8.0
1.0	XS	0.189	0.008	0.662	0.900	6.0
1.0	DBC	0.333	0.007	0.626	0.776	8.0
1.0	DBCL	0.190	0.008	0.673	0.897	8.0
1.0	XO	-0.021	0.013	0.413	0.950	4.8
2.0	ALL	0.634	0.006	0.320	0.268	8.0
2.0	XS	0.361	0.008	0.482	0.809	6.0
2.0	DBC	0.504	0.006	0.401	0.583	8.0
2.0	DBCL	0.361	0.007	0.480	0.812	8.0
2.0	XO	-0.007	0.013	0.411	0.944	4.8

As the true half-life rose from 0 to 2 weeks, the bias of XS rose. It was 0.005 at 0.2 week, 0.072 at 0.5 week, 0.189 at 1 week, and 0.361 at 2 weeks. Its MCSE was at most 0.008. The

coverage of XS fell to 0.90 at 1 week and 0.81 at 2 weeks. At a half-life of zero, XS had bias -0.016 with MCSE 0.008, where the closed-form bias is zero. When the true half-life was zero, DBC had bias -0.158 and coverage 0.92.

The power of DBC fell from 0.929 at 0.2 week to 0.401 at 2 weeks. The power of DBCL fell less, from 0.842 to 0.479. DBCL had higher power than DBC at 1 and 2 weeks. At 1 week its bias was 0.190, against 0.333 for DBC.

The oracle XO had absolute bias of at most 0.021 at every half-life. It analyzed 4.8 periods per participant at 1 week, and its power fell to 0.413 at 1 week and 0.411 at 2 weeks. Across the S1 cells, the simulated bias of every N-of-1 rule was within 2.4 MCSE of the closed form.

The paired difference in power between XS and RCT8_r08 at $\beta_D = -1$ and 0.2 week was 0.189, with paired MCSE 0.014. The difference between DBC and RCT8_r08 was 0.292 (MCSE 0.012). The difference between XS and DBC was -0.103. Its paired MCSE was 0.007, against 0.010 if the two arms had been treated as independent. At 2 weeks, XS had lower power than RCT8_r08 by 0.121 (MCSE 0.016).

4 Discussion

4.1 Principal findings

Under this linear DGP, power is a deterministic function of the two variance components, the lag-one correlation, the participant count, the design, and the carryover rule. The closed form reproduces the power of the stored v1 run to within 0.03. In the present run it agreed with the simulated power at $\beta_D = -1$ within 1.3 MCSE for every arm, and the Type I error stayed between 0.043 and 0.054. The standard error of the one-visit comparator is 2.31 times that of XS. Most of that ratio reflects the number of observations per participant. Giving the parallel-group trial eight visits and a covariate with reliability 0.8 reduces the ratio to 1.26.

The multi-crossover still has higher power than the matched-burden comparator at $\beta_D = -1$. The closed-form power is 0.84 for XS and 0.93 for DBC, against 0.63. In the simulation, the paired difference between XS and RCT8_r08 was 0.189 (MCSE 0.014). Two features of the DGP limit this comparison. It has no placebo response, so the blinded structure of the parallel-group trial earns no credit. It also has no period or order effect, so the randomized order of the N-of-1 arm is not tested.

4.2 Carryover

The schedule rule XS is nearly unbiased at the pharmacological half-life and biased at behavioral half-lives of 0.5 week or longer. In the simulation its bias was 0.005 at 0.2 week and 0.189 at 1 week. The rule DBC has the best power at 0.2 week, which is the half-life it assumes. Among the rules that handle carryover, it has the largest bias at one week and longer. The lag regressor in DBCL reduces that bias to the level of XS at a cost in standard error. Ignoring carryover altogether, as ALL does, biased the estimate toward zero even at

0.2 week and reduced coverage to 0.89 at $\beta_D = -1$. The oracle shows that unbiasedness at long half-lives requires dropping most periods. Our results on carryover depend on the DGP's assumption that carryover is an exponential decay of the on-drug effect. Other shapes were not simulated.

An earlier version of this paper reported that period exclusion was robust across half-lives. That result was an identity of the oracle rule, which dropped contaminated periods whatever the half-life. It is not a property of any rule that a trial can apply.

4.3 Comparison with prior studies

Blackston et al. found higher power for aggregated N-of-1 trials than for a parallel-group trial, and noted inflated Type I error under unmodeled carryover[5]. The present closed form agrees on direction. Our DGP does not reproduce their Type I inflation because carryover is proportional to β_D, so it vanishes under the null. A design with a carryover that is independent of the treatment effect would need a different DGP. Chen and Chen compared four analyses, including a mixed model with a carryover regressor[18]. The DBCL rule is of that kind. Tang and Landes show that single-participant tests need corrections for serial correlation[19]. In their Table 2, with four observations, the effect detectable with 80% power rises from 1.65σ at $\rho = 0$ to 4.08σ at $\rho = 0.4$ and 13.73σ at $\rho = 0.6$. Those values concern one participant and a one-sided 0.05 level. They do not apply to the aggregated analysis here, where 35 participants share the estimation of ρ.

4.4 Limitations

1. The DGP is linear with Gaussian errors. Nightmare counts are nonnegative integers.
2. There is no treatment-effect heterogeneity ($\tau^2 = 0$), so individual effects are undefined and the Senn trade-off between participants and cycles[13] is not examined.
3. There is no placebo response, natural-history trend, period effect, dropout, or missing data.
4. The design is a multi-crossover. It is not the Hybrid design.
5. The parameters are assumed. They are not estimated from prazosin trials.
6. The simulation has 2,000 replicates per cell. A difference in power of less than about 0.03 between two independent cells is within two MCSE. The closed form ignores the sampling error of the estimated variance components.
7. Both arms share one participant pool within a replicate, so arm comparisons within a cell are paired. Different cells use independent streams.

5 Conclusions

In a linear DGP at 35 participants, a randomized multi-crossover has higher power than a one-visit parallel-group trial mainly because it observes each participant several times. Matching the number of visits removes most of the standard-error gap. The remaining advantage

depends on the carryover rule. At a pharmacological half-life the schedule rule is nearly unbiased, and at behavioral half-lives of 0.5 week or longer every applicable rule is biased or loses power. A simulation with 2,000 replicates per cell agreed with the closed form within 1.3 MCSE for power at an effect of -1. The Type I error of every arm was between 0.043 and 0.054.

6 Glossary

- **ALL, XS, DBC, DBCL, XO.** The five N-of-1 analysis rules in Table 4.
- **Analysis half-life $t^a_{1/2}$.** The half-life the analysis assumes when it builds D_{bc}. The policy value is 0.2 week.
- **Carryover fraction.** The mean over a period of $2^{-s/t_{1/2}}$, the share of the on-drug effect that remains.
- **CR2.** The bias-reduced cluster-robust variance estimator with Satterthwaite degrees of freedom.
- **Containment df.** The between-within degrees of freedom that `nlme` reports for a within-participant term.
- **D_{bc}.** The drug indicator extended by the carryover fraction in off-drug periods after exposure.
- **Hybrid design.** Open-label titration, then blinded discontinuation, then a brief crossover. It is not simulated here.
- **Multi-crossover.** A randomized order of equal numbers of drug and placebo periods within each participant.
- **Oracle.** A rule that uses the true half-life. XO is the only oracle in this paper.
- **RCT1_exact, RCT1_r08, RCT8_r08.** The three parallel-group comparators defined in the Methods.
- **Reliability r_{BL}.** The share of the variance of the baseline covariate that is due to the true participant intercept.
- **$t_{1/2}$.** The carryover half-life of the DGP, in weeks.

7 Abbreviations

ADEMP, aims, data-generating mechanisms, estimands, methods, performance measures. AR(1), first-order autoregressive. BDC, blinded discontinuation. CAPS, Clinician-Administered PTSD Scale. DGP, data-generating process. GLS, generalized least squares. MCSE, Monte Carlo standard error. PTSD, post-traumatic stress disorder. RCT, randomized controlled trial. REML, restricted maximum likelihood.

8 Declarations

Ethics approval and consent to participate. Not applicable. No human participants were studied.

Consent for publication. Not applicable.

Availability of data and materials. The code is in the pmsimstats-ng repository under `analysis/scripts/treatment-main-effect/`. The closed-form results are computed when this document is rendered. Replicate-level simulation output is stored under `analysis/data/derived_data/04-mainfx-pilot/`, one file per cell, and the cell summary is `analysis/data/derived_data/04-mainfx-pilot-summary.rds`. The repository URL will be added at submission.

Competing interests. The authors declare no competing interests.

Funding. This work received no specific funding.

Authors' contributions. To be completed by the author team before submission.

Acknowledgements. None.

9 Reproducibility

The random-number scheme is one L'Ecuyer-CMRG chain seeded with `20261028`, advanced with `parallel::nextRNGStream()`. Replicate i of cell c uses stream number $(c-1) \times 2000 + i$ of that chain, and each replicate sets its stream explicitly. The production run stores one row per replicate and rule. The per-replicate streams can be regenerated from the seed with `make_streams()` in `pilot-core.R`. The stored run used R 4.6.0 on x86_64-pc-linux-gnu with 4 cores and finished on 2026-10-29 03:28 UTC. Simulation results depend on the platform's BLAS and LAPACK in the last digits.

The fast checks (`check-pilot.R`) cover the structure of the rules, the agreement of the closed form with the stored 2026-05-19 run, the agreement of the closed form with a 60-replicate simulation, and the reproducibility of the streams. They run in about 15 seconds. They do not replace the production run.

10 References

1. Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. *Journal of Clinical Epidemiology.* 1997;50(4):401-410. doi:10.1016/S0895-4356(96)00429-5

389 2. Guyatt G, Sackett D, Taylor DW, Chong J, Roberts R, Pugsley S. Determining optimal therapy: Randomized trials in individual patients. *New England Journal of Medicine*. 1986;314(14):889-892.

390 3. Duan N, Kravitz RL, Schmid CH. Single-patient (n-of-1) trials: A pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *Journal of Clinical Epidemiology*. 2013;66(8):S21-S28. doi:10.1016/j.jclinepi.2013.04.006

391 4. Schmid CH, Yang J. Bayesian models for N-of-1 trials. *Harvard Data Science Review*. 2022;(SI3). doi:10.1162/99608f92.3f1772ce

392 5. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of Aggregated N-of-1 Trials with Parallel and Crossover Randomized Controlled Trials Using Simulation Studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137

393 6. Carrozzo AE, Zimmermann G, Bathke AC, Neunhaeuserer D, Niebauer J, Kulnik ST. Two-arm crossover randomized controlled trial versus meta-analysis of n-of-1 studies: Comparison of statistical efficiency in determining an intervention effect. *Biometrical Journal*. 2025;67(2):e70045. doi:10.1002/bimj.70045

394 7. Raskind MA, Peskind ER, Kanter ED, et al. Reduction of nightmares and other PTSD symptoms in combat veterans by prazosin: A placebo-controlled study. *American Journal of Psychiatry*. 2003;160(2):371-373.

395 8. Raskind MA, Peterson K, Williams T, et al. A Trial of Prazosin for Combat Trauma PTSD With Nightmares in Active-Duty Soldiers Returned From Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133

396 9. Raskind MA, Peskind ER, Chow B, et al. Trial of Prazosin for Post-Traumatic Stress Disorder in Military Veterans. *New England Journal of Medicine*. 2018;378(6):507-517. doi:10.1056/NEJMoa1507598

397 10. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

398 11. Percha B, Baskerville EB, Johnson M, Dudley JT, Zimmerman N. Designing robust N-of-1 studies for precision medicine: Simulation study and design recommendations. *Journal of Medical Internet Research*. 2019;21(4):e12641. doi:10.2196/12641

399 12. Wang Y, Schork NJ. Power and Design Issues in Crossover-Based N-Of-1 Clinical Trials with Fixed Data Collection Periods. *Healthcare*. 2019;7(3):84. doi:10.3390/healthcare7030084

400 13. Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-383. doi:10.1177/0962280217726801

401 14. Yang J, Steingrimsson JA, Schmid CH. Sample size calculations for n-of-1 trials. Published online 2021.

402 15. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086

403 16. Pfizer. Minipress (prazosin hydrochloride) capsules: Prescribing information.

404 17. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business and Economic Statistics*. 2018;36(4):672-683. doi:10.1080/07350015.2016.1247004

405 18. Chen X, Chen P. A comparison of four methods for the analysis of N-of-1 trials. *PLoS ONE*. 2014;9(2):e87752. doi:10.1371/journal.pone.0087752

406 19. Tang J, Landes RD. Some t-tests for N-of-1 trials with serial correlation. *PLoS ONE*. 2020;15(2):e0228077. doi:10.1371/journal.pone.0228077